

(Quasi-)indexical tense: on temporal interpretation of Kazym Khanty embedded clauses

This paper describes a temporal system of Kazym Khanty and shows how non-finite tense is constrained in habituais and propositional attitudes. I suggest that emerging temporal patterns of Kazym Khanty can be explained by assuming that tenses function as logophoric pronouns, or indexiphors in terms of Deal (2018, 2020), and using introducing the anchor time as an interpretation parameter. This assumption is supported by observations that tenses differ from temporal indexical adverbials in their shifting behaviour and obey the Locality and Shift Together constraints.

Interpretation of embedded tense has been widely discussed in the existing semantic literature. Putting Sequence of Tense aside, it has been observed that tense in complements of propositional attitudes is interpreted not as absolute, but rather as relative. See a slightly modified example in Russian from (Grønn & von Stechow, 2010). Here the time of someone living in Moscow overlaps with his speaking time and does not necessarily overlap with time when the matrix clause was uttered.

- (1) On skazal, čto živet pod Moskvoj.
he say.PFV.PST that live.IPFV.PRES under Moscow
'He said he was living just outside Moscow.'

There are two primary theories which aim to derive the relative readings of embedded tense, which acts as deictic in unembedded environments. The first approach is a monster-based theory of Schlenker (1999, 2003). Schlenker suggests that tense is indexical, i.e. always interpreted in relation to the contextual Speech Time parameter. However, that parameter might be shifted by special context-altering operators called *monsters*.

The second approach centers on binding. It is proposed by von Stechow (2004) and implemented in several following works (Grønn & von Stechow, 2010, 2011). It assumes that tenses undergo variable binding under modals and attitudes, which are essentially quantifiers over times. There, characteristic features of tenses are deleted under agreement with a c-commanding modal and, consequently, not interpreted at LF. The same mechanism is used to derive logophoric pronouns that refer to individuals.

In this paper I examine embedded tense in the Kazym dialect of Northern Khanty (Uralic, Khantyic) and show that it posits certain difficulties for these theories of tense. Precisely, I will be focusing on non-finite temporal morphemes in gerunds, that appear in temporal adverbial clauses and sentential arguments. I show that tense in Kazym Khanty *when*-clauses is re-anchored in habitual contexts and argue that it cannot be attributed to binding. To account for

habitual anchoring and relative tense under attitudes, I suggest that Khanty temporal pronouns are not proper indexicals, but indexiphors in terms of Deal (2018). The similarities between tenses and logophoric pronouns have been observed before in (Stechow & Zimmermann, 2005; von Stechow, 2004). I suggest that an assumption that tense is indexiphoric provides two valuable insights. First, it explains why tenses are bound by habituais and not by other iterative quantifiers like *each of the three times*. Second, it underlines how patterns of temporal shift in grammatical tenses are completely different from the ones observed with temporal adverbials, assuming that tenses are indexiphors and temporal adverbials are proper indexicals. Altogether, this paper aims to explain a set of temporal phenomena in an underdescribed language with a single set of tools and collect the insights these data provide on existing approaches to embedded tense.

Section 1 contains the necessary details of Kazym Khanty grammar, together with methodological remarks and initial assumptions about tense. Section 2 discusses *when*-clauses and introduces a notion of habitual anchoring. In Section 3, I show why existing approaches to embedded tense are unable to predict Khanty data and implement an indexiphor-based approach to account for habitual anchoring. In Section 4, I sketch out the formal implementation of the analysis. Section 5 discusses problematic data and concludes.

1. Preliminaries

1.1. Background on Khanty and methodology

Kazym Khanty is a dialect of Northern Khanty, which is a Uralic language that was spoken by around 1700 people at the time of writing of (Kaksin, 2007). Kazym Khanty is predominantly located in the Kazym village of the Khanti-Mansi Autonomous District of Russia and in other smaller settlements situated around the Kazym river. Kazym Khanty is not passed down to children anymore and is therefore endangered (Aristova, 2023).

Khanty is an SOV order language. Khanty verbs have two finite tenses – past ($-(\partial)s$, PST) and non-past ($-(\partial)\lambda$, NPST). The same also applies for non-finite forms, which exhibit non-finite non-past ($-ti$, NFIN.NPST) and non-finite past ($-\partial m$, NFIN.PST) tenses. Although I generally call those non-finite forms tenses, following the customary notation, they are most likely located in the projections of ordering aspect, see <anonymized details>. Besides that, there are no dedicated markers of ordering and viewpoint aspect in Khanty. Perfect-like and indirect evidential meanings are sometimes conveyed with insubordinated non-finite forms with possessive agreement. While finite verbs have a dedicated set of verbal agreement markers, possessives mark agreement on the non-finite forms. Descriptions of Kazym Khanty grammar can be found in (Kaksin, 2007, 2008; Nikolaeva, 1999).

Khanty does not show Sequence of Tense phenomena with finite or with non-finite tenses. See (2), where past tense marking on the complement cannot have a simultaneous interpretation and is strictly *de se*. Similarly, in (3) a PST-marked complement is shifted backwards relative to the time of saying.

- (2) // I called Masha and asked her what Vasya is doing//
 mašaj-en lup-əs wašaj-en uλ / uλ-t-aλ /
 m.-POSS.2SG say-PST[3SG] V.-POSS.2SG sleep-NPST[3SG] / sleep-NFIN.NPST-POSS.3SG/
 #uλ-əs / #uλ-m-aλ
 #sleep.PST / #sleep-NFIN.PST-POSS.3SG
 ‘Masha said that Vasya slept (when I called)’
- (3) //Today I called Masha and asked her what Vasya has been doing during weekends//
 mašaj-en lup-əs wašaj-en uλ-əs / uλ-m-aλ /
 m.-POSS.2SG say-PST[3SG] V.-POSS.2SG sleep-PST[3SG] / sleep-NFIN.PST-POSS.3SG /
 #uλ / #uλ-t-aλ
 #sleep.NPST / #sleep-NFIN.NPST-POSS.3SG
 ‘Masha said that Vasya was sleeping (on the weekends)’

All glossing lines are be formatted according to Leipzig Glossing Rules. Glosses will be listed in Appendix I. Descriptions of Northern Khanty grammar can be found in (Nikolaeva, 1999) and in (Kaksin, 2007, 2008) specifically for the Kazym dialect.

The data used in this study were collected during three consecutive fieldtrips to Kazym village, <anonymized details>. I worked with a total of 10 native speakers and I am grateful to each and every one of them for their patience and eagerness to teach. All of the examples used in this study were collected by me and verified with three or more speakers. Here I will use Khanty instead of Kazym Khanty for reasons of space. Any time that I mention Khanty, it is Kazym Khanty that I am referring to.

All examples below are mine, unless mentioned otherwise. The elicitation process was organized following the basic principles of semantic fieldwork (Kibrik, 2005; Matthewson, 2004). Contexts were provided in two ways. Some of them were delivered in Russian, and such contexts will be shown in curly brackets as {...Context...}. In other cases, the consultants were asked to translate a short dialogue from Russian to Khanty sentence-by-sentence and the surrounding sentences functioned as a more-or-less natural context for the target sentence. Such dialogue-framed contexts will be framed by double slashes as in //...Context...//. This method of elicitation was mainly inspired by the notion of narrative elicitation (Burton & Matthewson, 2015; Louie, 2015) and proved to be quite effective when working with tense. I refrained from using regular narratives to avoid phenomena associated with narrative progression. Narratives allow for anaphoric interpretations of tense, which shift the evaluation time from the utterance time to a time of the story (Schiffrin, 1981). Presented below is a

fragment of a fairytale taken from the corpora of texts collected during <anonymized details>. Note how the temporal marking changes from past to non-past throughout the story. Such narrative-like phenomena complicate the semantic investigation into embedded tense to a noticeable extent. To make sure that narrative progression would not influence the interpretation of embedded tense, narratives were avoided altogether.

- (4) in jiw puχ-j-ən wan-a tuχi śi χāt-əλ
this bear-DIM-POSS.2SG short-DAT there EMPH move-NPST[3SG]
'The Bear came closer'
- (5) t̪əm sort-əλ k̪uś kunś-əmə-s-λə
that pike-POSS.3SG though catch-MOM-PST-3SG>SG
'Almost caught the pike'
- (6) kunś-əmə-ti w̪uś-ijəλ-s-λe katλn-əλ-ən
catch-MOM-NFIN.NPST want-FREQ-PST-3SG>SG bear.claws-POSS.3SG-LOC
'He wanted to catch her with his claws.'
- (7) in sort-en śur-əmt-əs
this pike-POSS.2SG leap-PUNCT-PST[3SG]
'The Pike leaped out,'
- (8) jink-a iλ m̪ərət-əs w̪əri k̪utəp-ən nuχ et-əs
water-DAT down dive-PST[3SG] dead.channel middle-LOC rise-PST[3SG]
'Dived down, emerged in the middle of a still channel'
- (9) juχli p̪əla want-əλ
back toward look-NPST[3SG]
'(And) looked back' (<anonymized details>)

1.2. Initial assumptions on tense

While describing the data, the following abbreviations will be used:

ST — Speech Time;

MT — time of event described in the main clause;

ET — time of event described in the embedded clause.

For example, in a sentence *Maria found out that kids stole her slippers*, MT is the time of Maria finding out and ET is the time of kids stealing slippers.

This paper is mainly concerned with interpretations of non-finite tense in gerunds that function as temporal adverbial clauses and sentential arguments. Gerunds have been argued to project up to CP (Bikina et al., 2022) and I argue that the amount of structure they contain allows them to acquire absolute interpretations of tense. Non-finite tense also appears in other, structurally simpler embedded clauses, and it was shown that non-finite tense in AspP-level relative clauses is always interpreted as relative to the nearest temporal variable (Sidorkina, 2024).

Since the non-finite tense marking in RCs is morphologically equivalent to the non-finite tense in gerunds, it seems reasonable to maintain an initial assumption of monosemy. Therefore, I assume the following: non-finite tenses in Khanty are existential quantifiers over the domain of time intervals. Denotations of non-finite finite tenses are shown in (10-11), where $\geq$ and $<$ are relations on an ordered set of time intervals, as defined in (12). Although the domain for quantifiers in these temporal operators needs to be contextually restricted to derive the necessary interpretations, I will ignore the mechanisms of contextual restriction here for reasons of space.

$$(10) \llbracket \text{NFIN.NPST} \rrbracket^{c,g} = \lambda p \in D_{\langle i, t \rangle}. \lambda t \in D_i. \exists t'. [t' \geq t \wedge P(t')]$$

$$(11) \llbracket \text{NFIN.PST} \rrbracket^{c,g} = \lambda p \in D_{\langle i, t \rangle}. \lambda t \in D_i \exists t'. [t' < t \wedge P(t')]$$

$$(12) \begin{aligned} t \geq t' & \text{ iff the left boundary of } t \text{ is later or equal to the left boundary of } t' \\ t < t' & \text{ iff the right boundary of } t \text{ is earlier than the left boundary of } t' \end{aligned}$$

Since NFIN.(N)PST markers can appear in RCs, which lack temporal projections, I will place the corresponding operators into the domain of ordering aspect. However, here I will continue calling them tenses, because they operate with temporal variables and, in absence of proper temporal operators in gerunds, they bear the functions of tense, i.e. locate the time of the preajacent event in relation to ST.

Following (Grønn & von Stechow, 2010; von Stechow, 2009) I assume that the architecture of the temporal domain consists of two components – temporal anchors and temporal shifters. Temporal shifters are quantifiers that locate the event time of the preajacent in a certain way in relation to the temporal anchor. Temporal anchors, in turn, are pronouns that serve as evaluation times for the clause. I suggest that Khanty employs two kinds of the temporal anchors introduced in the original works. First is a deictic pronoun t^* that refers to the Speech Time (ST). Second is an empty pronoun TPRO that moves out and leaves behind a lambda-abstructor. Consequently, clauses with TPRO contain tense that is always interpreted as relative. The resulting inventory is shown in Figure 1. TPRO is introduced in the syntax, as to fill the temporal argument, however, clauses with TPRO simply translate to unsaturated predicates of times in semantics.

T-center: t^* , TPRO

T-shifter: NFIN.NPST, NFIN.PST

Figure 1. Khanty temporal inventory

Here I assume that the temporal anchor of the clause is determined by amount of syntactic structure. In Khanty, there are AspP-level non-finites, such as relative clauses, which are interpreted relative to the matrix clause time and, supposedly, contain the TPRO anchor, see (Sidorkina, 2024). Here, we will focus on higher-projecting structures, namely, CP-level

gerunds. As gerunds project up to CP, there is enough space for a t^* anchor and gerunds are predicted to have an absolute temporal interpretation. In the following sections I will show that this is not always the case and suggest that the necessary predictions can be obtained if we question the indexicality of t^* .

2. Temporal adverbial clauses

In this Section I will discuss Khanty *when*-clauses and introduce a puzzle of habitual anchoring. Then, I will proceed to build a baseline architecture of *when*-clauses and show why the account of temporal binding cannot straightforwardly predict the temporal patterns observed in Khanty.

2.1. Introducing LOC-adverbs

Khanty *when*-clauses appear in form of locative-marked gerunds, i.e. as non-finite verb forms with obligatory possessive subject agreement, see (13). Here I will call such temporal adverbial clauses LOC-adverbs.

- (13) stem-NFIN.(N)PST-POSS-LOC
'When STEM, ...'

In the previous section it was assumed that gerunds contain the deictic temporal anchor t^* . In line with that assumption, LOC-adverbs contain absolute tense. In example (14), where the event of entering the room precedes the speech time, NFIN.PST is felicitous and NFIN.NPST is not. One may suspect that NFIN.PST in (14) is licensed by a perfective aspect in the embedded clause. However, it is important to note that the infelicity of NFIN.NPST in (14) cannot be attributed to perfectivity, because it is not available even in clearly imperfective clauses with past temporal reference, like (15-16).

- (14) $\mu\lambda\chi\alpha t\acute{\alpha}\lambda$ [ma $\lambda\upsilon\eta\acute{o}t$ -{**m/#t**}- ϵm - $\acute{\alpha}n$] $\alpha\rho\acute{s}\epsilon$ - m $\upsilon\lambda$ - $\acute{\alpha}s$
yesterday [I enter-{**NFIN.PST/#NFIN.NPST**}-POSS.1SG-LOC] sibling-POSS.1SG sleep-PST[3SG]
‘Yesterday, when I entered (the room), my brother was sleeping’
- (15) $^{\circ}k\mu\lambda\chi\alpha t\acute{\alpha}\lambda$ [ma $\lambda\upsilon\eta\acute{o}t$ -{**m/#t**}- ϵm - $\acute{\alpha}n$] $\alpha\rho\acute{s}$ - ϵm $\upsilon\lambda$ - $\acute{\alpha}s$
yesterday [I read-{**NFIN.PST/*NFIN.NPST**}-POSS.1SG-LOC] father-POSS.1SG sleep-PST[3SG]
‘Yesterday, when I was reading, my father slept’
Comment on $\# \lambda\upsilon\eta\acute{o}t$ - t - ϵm - $\acute{\alpha}n$: ‘it means he is reading now’
- (16) [ma $\upsilon\lambda$ -**m/#t**- ϵm - $\acute{\alpha}n$] $\rho\epsilon t$ ’a-jen¹ ow - ϵm $\sigma\epsilon\eta k$ - $\acute{\alpha}s$
I sleep-{**NFIN.PST/#NFIN.NPST**}-1SG-LOC Petya-POSS.2SG door-POSS.1SG knock-PST[3SG]
‘When I was sleeping, Petya knocked at my door’. (Muravyev 2019, modified)
- MT = ET ----- ST --->

¹POSS.2SG and POSS.3SG may function as determiners without signaling proper possessive relations, see (Mikhailov, 2023, 2024).

Here I will be using schemas, where green arrows represent temporal markers that are interpreted as relative tense and red arrows represent markers that are interpreted as absolute tense.

Past temporal reference blocks NFIN.NPST regardless of viewpoint aspect in the embedded clause. Therefore, the examples above show that non-finite tense in LOC-adverbs should primarily be regarded as a temporal marker rather than a marker of viewpoint aspect. Moreover, they demonstrate that non-finite tense in LOC-adverbs is not relative, as temporal overlap between the event of the LOC-adverb and the matrix clause does not license NFIN.NPST. In example (15), the time of reading coincides with sleeping and, if non-finite tense were relative, we would expect NFIN.NPST to be felicitous when it clearly isn't. In a similar fashion, in example (16) the time of sleeping overlaps with knocking and NFIN.NPST is not licensed. This evidence suggests that non-finite tense in LOC-adverbs is not viewpoint aspect and is not relative tense.

For future temporal reference, consider (17), where the event of finishing work is located in the future in relation to ST. As to expect, NFIN.NPST is felicitous and NFIN.PST is not. Here, once again, presence of NFIN.NPST cannot be attributed to aspect, as the embedded clause is clearly perfective. NFIN.NPST in such clauses can only be attributed to the future temporal reference in relation to ST. Therefore, we can safely assume that non-finite tense in LOC-adverbs is deictic and, as such, it relates ST and ET.

- (17) [puχ-εm jetn rəpət-ti jetsəptə-**{#m/t}**-aλ-ən]
 son-POSS.1SG evening work-NFIN.NPST finish-**{#NFIN.PST/ NFIN.NPST }**-POSS.3SG-LOC
 lapət-λ-εm
 feed-NPST-1SG>SG
 'When my son finishes work at the evening, I will feed him'

- (18) [λuw juχət-**t**-aλ-ən] nəm χuja-n ǎn lawəλ-λ-a
 [(s)he come-NFIN.NPST-POSS.3SG-LOC] nobody-LOC NEG wait-NPST-PASS
 'When he comes home, nobody will welcome him'
- ST ----- ET ----- MT --->

However, there is a certain environment where LOC-adverbs do not straightforwardly concur with this assumption. In habitual contexts LOC-adverbs obligatorily acquire NFIN.NPST marking. As an illustration, compare examples (19) and (20). Although both of them refer to event that took place before ST, one requires NFIN.PST marking on the LOC-adverb and the other — NFIN.NPST.

- (19) μυλχătəλ ma juχət-**{m/#t}**-εm-ən puχ-εm rəpit-əs
 yesterday I come-**{NFIN.PST/#NFIN.NPST}**-POSS.1SG-LOC son-POSS.1SG work-PST[3SG]
 saraj-ən

shed-LOC

‘When I came yesterday, my son was working in the shed’

- (20) kašəŋ χātəλ ma juχət-**{t/#m}**-εm-ən puχ-εm saraj-ən
every day I come-**{NFIN.NPST/#NFIN.PST}**-POSS.1SG-LOC son-POSS.1SG shed-LOC
rəpit-əs
work-PST[3SG]

‘Every day, when I came, my son would work in the shed’

Non-past marking in habituals is not caused by the time of the habit extending to ST. See (21), where the regular event of Speaker coming home and her dog waiting for her at home does not hold at ST, since the dog has grown old.

- (21) ma juχi juχət-**{*m/t}**-εm-ən amp-εm ma χuś-am-a
I home come-**{*NFIN.PST/NFIN.NPST}**-POSS.1SG-LOC dog-POSS.1SG I toward-POSS.1SG-DAT
isa jeśaλt χəχλ-əs
always to run-PST[3SG]
‘When I came back home, our dog would always wait for me’
//Now she is very old and doesn’t wait for anyone//

Hereafter I will refer to the unusual properties of non-finite tense which show up in habitual contexts as **habitual anchoring**, due to that the time of habitus in such cases serves as a T-anchor for temporal adverbials. After building up the baseline LF of LOC-adverbs in the following section, I will argue that the structure that I propose for LOC-adverbs is better suited to account for habitual anchoring than pre-existing analogous analyses (Arregui & Kusumoto, 1998), as it maintains the paradigmatic parity between the NFIN.PST and NFIN.NPST and better explains why certain temporal clauses do undergo habitual anchoring and others do not.

2.2. Constructing a when-clause

Arregui and Kusumoto (1998) together with Kusumoto (2017) suggest that there are generally two types of temporal clauses — the ones with independent temporal reference, which we will call **full-tensed** clauses, and the ones without reference of their own, which I will call **empty-tensed** clauses.

Empty-tensed clauses themselves are not propositions, but rather predicates of times of type $\langle i, t \rangle$. Assuming the architecture we introduced above, these clauses have a TPRO as a temporal anchor. An empty-tensed clause, here exemplified by English gerund, would have the denotation like the one in (23). Such temporal clauses lack a deictic T-anchor and are fully dependent on local evaluation times.

- (22) John_i left before [PRO_i saying good-bye]
(23) $\llbracket \text{PRO}_i \text{ saying good-bye} \rrbracket = \lambda t \in D_i. [\text{John says good-bye at } t]$ (Kusumoto, 2017)

Temporal connectives have the same semantics regardless of whether their complement is a fully-tensed or an empty-tensed clause. Here we will use simplistic denotations for Khanty

temporal connectives shown in (24). See (Beaver & Condoravdi, 2003; Ogihara, 2022; Rett, 2020) for a more sophisticated analysis of connectors.

- (24) $\llbracket j\ddot{u}prij\ddot{a}n \text{ (after)} \rrbracket = \lambda P \in D_{\langle i, t \rangle}. \lambda t \in D_i. \exists t' [P(t') \ \& \ t' < t]$
 $\llbracket LOC^2 \text{ (when)} \rrbracket = \lambda P \in D_{\langle i, t \rangle}. \lambda t \in D_i. P(t)$

Temporal adverbial clauses with empty-tensed clauses exhibit relative tense, since the temporal quantifier in those is evaluated not in relation to an independent temporal anchor, but in relation to the temporal anchor of the main clause.

Full-tensed clauses are inherently propositions which contain a deictic pronoun t^* as a temporal anchor. The pronoun saturates the temporal argument, which is open in empty-tensed clauses. However, in order to serve as an argument to a temporal connector of type $\langle it, it \rangle$, full-tensed clauses should somehow change their type to predicates of times. Geis (1970) and Larson (1990) originally suggested that temporal adverbial clauses are relative clauses with a relativized temporal argument. The corresponding decomposition of a *when*-clause is shown in (25).

- (25) a. when Vasya discovered America
 == at t_j such that Vasya discovered America at t_j
 b. before Vasya discovered America
 == before t_j such that Vasya discovered America at t_j

This idea is supported by movement-driven ambiguities known as Geis-ambiguities. See (26), where two interpretations are possible. Interpretation (a) involves movement of the relativized temporal argument t from the clause [you said at t]. Interpretation (b) involves long-distance movement from the embedded [you said [he had left at t]].

- (26) I encountered Satoshi in Amherst when you said he had left
 a. I encountered Satoshi in Amherst at a time which_j you said at t_j [he had left]
 b. I encountered Satoshi in Amherst at a time which_j you said [he had left at t_j]
 (Arregui & Kusumoto, 1998)

Arregui and Kusumoto (1998) and Kusumoto (2005, 2017) show that analogous ambiguities are characteristic of full-tensed temporal clauses and do not arise with empty-tensed ones, since empty-tensed clauses are themselves predicates of times and do not require relativization to change types. The resulting combination of phenomena characteristic of empty-tensed and full-tensed temporal clauses is shown below.

Type	Tense	Geis-ambiguities
Empty-tensed	Relative tense	-

² Note that LOC-adverbs do not precisely mean ‘at the interval t at which P took place’ and have additional (non-)overlap and evidential components (Muravyev, 2022), which can be conveniently incorporated into the semantics of a LOC modifier if need be.

Full-tensed	Absolute tense	+
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Table 1. Features of temporal clauses

Hence, to safely determine whether Khanty LOC-adverbs contain fully-tensed or empty-tensed clauses, we are to show that they align with either empty-tensed or full-tensed clauses on both parameters. As was shown above, tense in LOC-adverbs is absolute, making them alike to full-tensed clauses. However, as for Geis-ambiguities, those fail to act as a meaningful test for temporal relativization in Khanty for a number of reasons. First, they are extremely hard to elicit in a fieldwork setting. Consultants often reject sentences which would allow for a double reading for pragmatic reasons or paraphrase them, motivating it by statements like ‘Why would you use a giant sentence like that?’. Forcing the necessary configuration seemed unreasonable, as it would not yield genuine judgements, which we strive for throughout our elicitation sessions. Second, the interpretation that requires long-distance movement is expected to lack in Khanty due to independent island constraints³. Gerunds that function as sentential arguments strongly disallow *wh*-extraction (Bikina et al., 2022). Hence, the relativized temporal argument cannot move out from the complement of the verb *say* into the matrix clause.

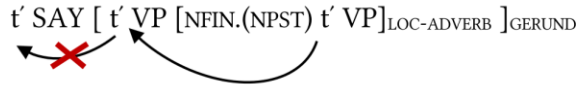
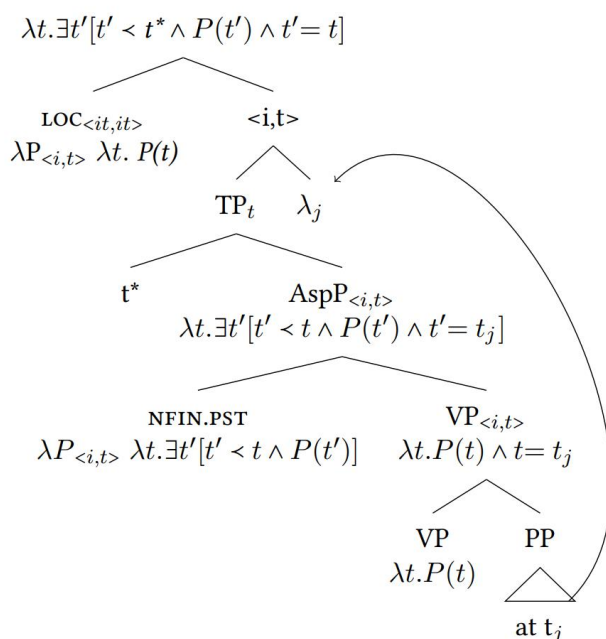


Figure 2. No long-distance temporal relativization from gerunds

That renders Geis-ambiguities an invalid test in case of Khanty. Consequently, additional evidence must be brought up in order to determine the inner makeup of LOC-adverbs. Such evidence may come from syntactic structure. LOC-adverbs project up to CP (Bikina et al., 2022), and it seems plausible to assume that they are propositions and not predicates of times. With that in mind, absolute tense in LOC-adverbs and their apparent proposition status provide sufficient ground to conclude that LOC-adverbs contain full-tensed clauses with a relativized temporal argument. The resulting structure is shown in Figure 3. The temporal variable moves from a complement of PP, leaving behind a trace t_j of type i , which is bound by a c-commanding lambda abstractor above CP by Predicate Abstraction (Heim & Kratzer, 1998), as in regular relative clauses. The resulting predicate of times is then modified by LOC, which for the purposes of the current study we assume to be a vacuous modifier.

³ <acknowledgement>



The resulting configuration allows us to predict the absolute tense in LOC-adverbs, since t^* normally refers to ST. However, unlike absolute tense, habitual anchoring phenomena does not obviously follow from the LF above. The next section explores how habitual anchoring could be derived from the proposed structure of LOC-adverbs.

2.3. Arregui and Kusumoto's analysis

Arregui and Kusumoto (1998) (A&K) discuss Japanese *when*-clauses and describe a pattern similar to the one we find in Khanty. Consider examples (27-28), both of which describe events in the past. A *when*-clause with past tense, as in (27), receives an episodic interpretation and a *when*-clause in present tense, as in (28), receives a habitual interpretation, as if Satoshi was coming multiple times.

- (27) [Satoshi-ga kita toki] Junko-wa heya-ni ita
[S-NOM come-PAST when] J-TOP room-at be-PAST
'Junko was in her room when Satoshi came' (episodic) (Arregui & Kusumoto 2015)
- (28) [Satoshi-ga kuru toki] Junko-wa heya-ni ita
[S-NOM come-PRES when] J-TOP room-at be-PAST
'Junko was in her room whenever Satoshi came' (habitual) (Arregui & Kusumoto 2015)

To explain such a pattern, A&K suggest that present tense in Japanese is not a quantifier, like past tense, but a bound variable instead. Therefore, they suggest, it can be bound by a habitual quantifier and past tense cannot. This analysis assumes that habituais contain a covert quantifier over times, which is located in VP. The quantifier receives a predicate of times as a restrictor and ranges over predicates of times. Since the quantifier has an additional temporal argument of its own to determine when the habit took place, the

resulting type is $\langle it, \langle it, it \rangle \rangle$. A&K do not discuss how the restrictor is saturated in the absence of over temporal adverbs and I assume that it is restricted covertly in such cases. Note that the precise adjunction site of a habitual quantifier may be debated, see (Boneh & Doron, 2008; Cable, 2020; Scheiner, 2002).

A&K's binding configuration for habitually anchored *when*-clauses is shown below in Figure 4. A&K argue that Japanese *when*-clauses have absolute tense, hence they contain a t^* as their temporal anchor and a relativized temporal argument, judging by the presence of Geis-ambiguities. The past tense in temporal adverbs is quantificational. That is precisely the system we adopted for Khanty. However, A&K suggest that present tense is not a quantifier like its PST counterpart, but rather a bound variable without proper quantificational force. Following from that, they claim that present tense can be bound by a habitual quantifier and past fails to do so. Consequently, past-marked *when*-clauses only give rise to episodic interpretations.

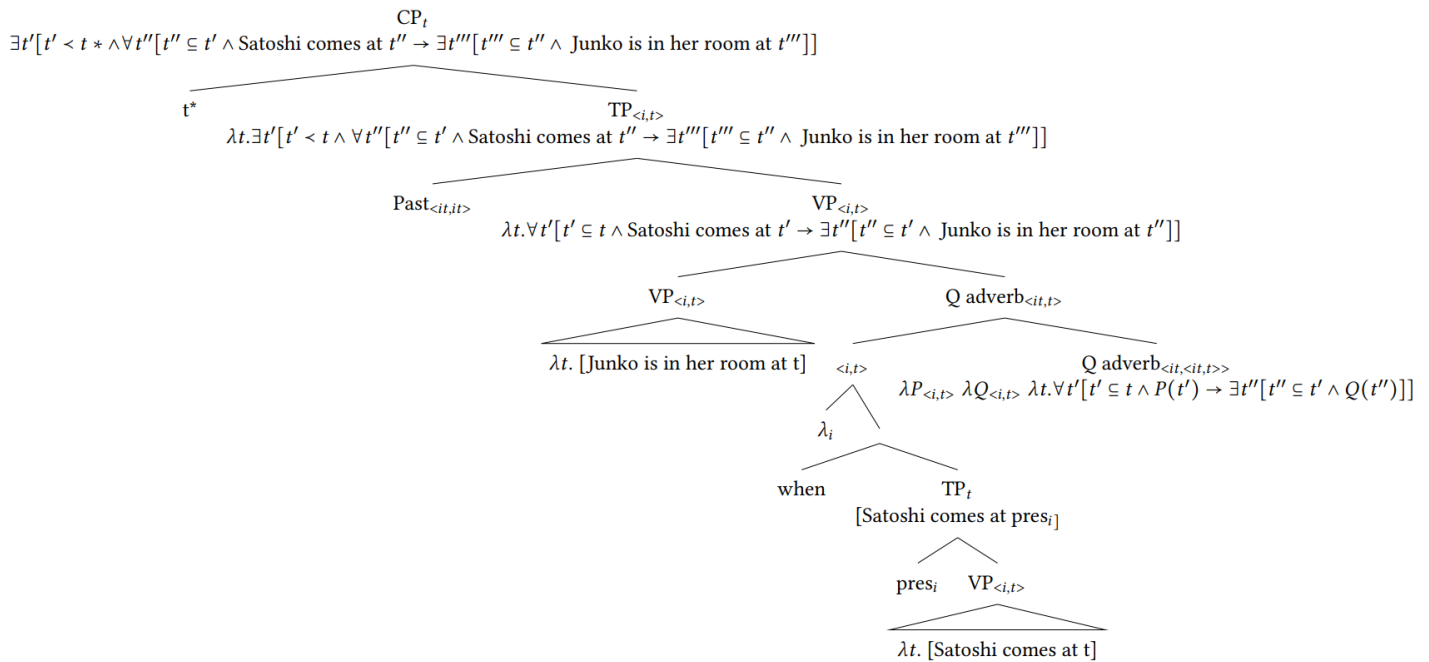


Figure 4. A&K's account of habitual anchoring in Japanese *when*-clauses

Now, note that in this structure the temporal anchor t^* is absent in the *when*-clause. If it were present, it would yield a type mismatch, as t^* of type i would not have been able to combine with a TP of type t , which is already saturated by the *pres*. Therefore, within A&K's system, clauses with present tense marking have the same structure regardless of whether they are empty-tensed or full-tensed. Another important consequence that comes with the assumption that present tense is a variable and not a quantifier, is structural irregularity between present and past marking, which might have unwanted consequences. For example, if present and past are not structural alternatives, then we expect them to not enter pragmatic

competition, which might influence cessation inferences and similar phenomena (Altshuler & Schwarzschild, 2013; Cable, 2017). Moreover, as will be shown below, while being fine with full-tensed *when*-clauses, this account fails with full-tensed *after*-clauses.

Japanese *after*-clauses are empty tensed and do not undergo habitual anchoring, see (29).

- (29) {I used to have a couple of colleagues at our company, Yamada and Sato. Both of them don't work anymore. It's quite peculiar that I have never seen them working the same shift}⁴
 satou-san ga kaet-ta / *kuru ato de yamada-san wa ki-ta
 Satou-HON NOM return-PST / *return-PRES after LOC yamada-HON TOP come-PST
 'Yamada (always) came after Sato returned home'

Khanty *after*-clauses are a mirror image of those, as they contain absolute tense and are habitually anchored, just as Khanty and Japanese *when*-clauses. Examples (30-31) show that tense in Khanty *after*-clauses, which consist of a gerund and a postposition *jupijən*, is interpreted in relation to the ST and not MT.

- (30) [waśaj-en tut jux sewr-əm / #sewər-ti **jupijən**] mən kər-əw
 V.-POSS.2SG fire tree chop-NFIN.PST / #chop-NFIN.NPST after we stove-POSS.1SG
 ăł-s-əw
 heat-PST-1PL
 'After Vasya chopped the wood, we laid a fire'

ET ----- MT ----- ST --->

- (31) [waśaj-ən tut jux sewər-{t/#m}-aλ **jupijən**]
 V.-POSS.2SG fire tree chop-{NFIN.NPST/#NFIN.PST}-POSS.3SG after
 mən kər-əwăł-λ-əw
 we stove-POSS.1SG heat-NPST-1PL
 'After Vasya chops the wood, we will lay a fire'
 Comment on #sewər-m-aλ 'chop-NFIN.PST-POSS.3SG': 'It means he has chopped it already'

ST ----- ET ----- MT --->

Example (32) shows that Khanty *after*-clauses are habitually anchored, since they are marked with NFIN.NPST despite the fact that sister's regular arrivals from the city are located in the past.

- (32) up-əm isa [woš-ewełt juxət-{t/#m}-aλ **jupij-ən**] isa
 sister-POSS.1SG always [town-from come-{NFIN.NPST/#NFIN.PST}-POSS.3SG after] always
 jăŋx-əs wənt-a
 go-PST[3SG] forest-DAT
 'After coming home from the city my sister would always go to the forest'
 //But now she moved to the city for good//

⁴ I thank Anna Kulikova for helping me to obtain the data.

Let us try and apply A&K's analysis to example (32). Since habitual anchoring forces the embedded clause to acquire NFIN.NPST marking, we will follow A&K's line of thought and assume that Khanty NFIN.NPST is also a bound variable and not a quantifier. Note that, unlike *when*-clauses, where the temporal connective may be regarded as a vacuous relativization marker, *after*-clauses contain a non-vacuous temporal quantifier. This is problematic for a bound NFIN.NPST analysis, since it results in a type mismatch, see below. The non-past tense saturates the temporal argument and the temporal connective *after* cannot modify anything that is not a predicate of times.

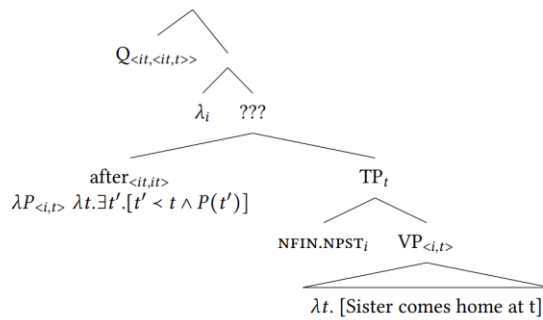


Figure 5. *Type mismatch in habitual anchoring of after-clauses*

If the abstraction happens not in the immediate scope of the quantifier, but below the temporal connective, there is no type mismatch and the derivation proceeds as shown below in Figure 6, yielding intuitively correct truth-conditions.

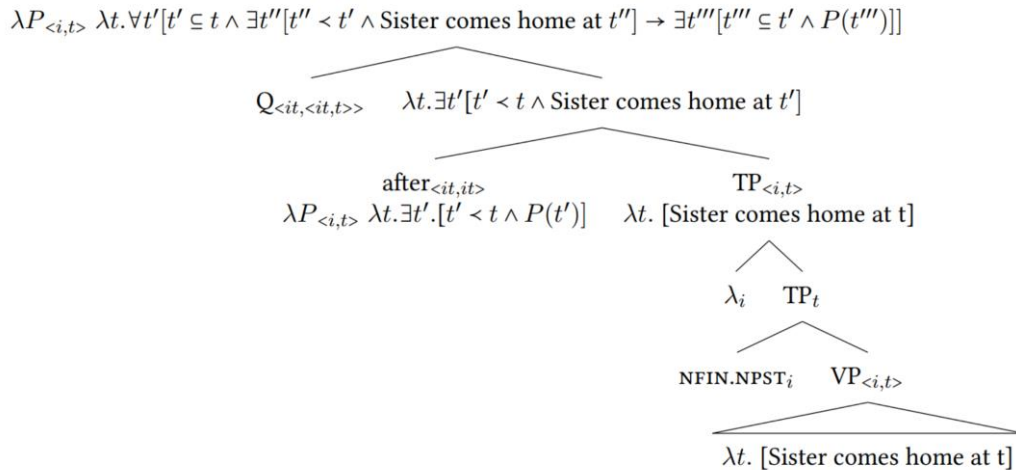


Figure 6. *Habitual anchoring of after-clauses within A&K's analysis*

However, this structure in no way depends on the properties of the embedded clause – namely, whether it is fully-tensed or empty-tensed. The binding follows not from the presence of a t^* anchor, but from the properties of non-past or present tense. If the structure like the one

in Figure 6 is possible for fully-tensed *after*-clauses like Khanty ones, there is nothing precluding it from being possible for Japanese *after*-clauses, which are empty-tensed. This is problematic, since, as was shown above, while Khanty *after*-clauses undergo habitual anchoring, Japanese clauses do not. Japanese *after*-clauses in habitual environments disallow present tense and yet the structure akin to one in Figure 6 does not exclude the availability of present tense. In fact, it shows that present tenses should be available in *after*-clauses regardless of their temporal anchor, TPRO or t^* .

A&K analysis does not account for the observed parallels between the presence of a t^* anchor in a temporal clause and habitual anchoring (see Table 2).

After-clauses	Japanese	Khanty
Temporal reference with past tense?	relative	absolute
T-anchor (with past tense)	TPRO	t^*
Habitually anchored?	no	yes

Table 2. *Japanese vs Khanty after-clauses*

The following pattern seems to arise: temporal clauses with a deictic t^* anchor do undergo habitual anchoring and clauses with a TPRO anchor do not. A&K's analysis fails to account for this pattern. Therefore, I suggest that habitual anchoring in both Khanty and Japanese cannot be attributed to temporal operators themselves, but rather it should somehow depend on the indexical qualities of a temporal anchor in the temporal clause.

2.4. Why temporal binding is not enough

If we approach habitual anchoring as t^* -anchoring, we can say that t^* refers to ST in episodic environments and shifts in habituals. Given that the most simplistic approach to habituality is to regard it as a universal quantifier, it might seem like the observed properties of t^* can be derived with a straightforward binding-based approach in spirit of (Grønn & von Stechow, 2011; Kusumoto, 2005; von Stechow, 2004). In this section I review this approach and show that it is not sufficient to account for Khanty data.

Can t^* be bound, whilst being an indexical? Underlining an analogy between the nominal and the temporal domain, (Kratzer, 1998) notes that deictic pronouns like *I* can be bound by quantifiers, loosing parts of their indexical meaning. See example (33), originally attributed to Irene Heim. One available interpretation is that nobody else got a question that I understood. Another interpretation is that nobody else got a question they understood. On the latter

interpretation, where the pronoun is bound, the person features on *I* are not interpreted in the presupposed component of meaning.

- (33) Only I got a question that I understood.

Presupposed: I got a question that I understood

Asserted: Nobody else got a question that I understood [1SG]

Nobody else got a question that (s)he understood. [1SG]

Similar examples were suggested by (Rullmann, 2004), cf. (34). Note that in (34a) bound *we* can lose its number features, though the free pronoun in (34b) cannot.

- (34) a. We each/all think we're the smartest person in the world. [PL]

b. #We're the smartest person in the world. [PL] (Rullmann, 2004)

Taking that into account, it is plausible to assume that a deictic pronoun t^* can be bound by a c-commanding quantifier and lose some of its features in the process, namely the feature stating that the pronoun refers to ST. Suppose that the habitual operator is a simple quantifier over times, as was suggested by A&K (1998), and it takes temporal adverbs as restrictors. Suppose that this quantifier binds t^* . Then, for the sentence (32), repeated below as (35), we acquire the meaning given in (36). The resulting denotation requires t' to be simultaneously earlier and not-earlier than the time of sister coming home. Therefore, the restrictor of the HAB quantifier is empty, which makes the formula trivially true.

- (35) up-εm isa [woš-eweλt juχət-{**t/#m**}-aλ **jupij-ən]** isa
 sister-POSS.1SG always[town-from come-{NFIN.NPST/#NFIN.PST}-POSS.3SG after] always
 jǎŋχ-əs went-a
 go-PST[3SG] forest-DAT
 ‘After coming home from the city my sister would always go to the forest’
 //But now she moved to the city for good//

- (36) $\llbracket \text{HAB [after coming from the city]} \rrbracket = \lambda P_{\langle i, t \rangle} \lambda t. \forall t' [\exists t'' [t'' \prec t' \wedge \exists t''' [t''' \succcurlyeq t' (\text{bound } t^*) \wedge \text{sister comes from city at } t''' \wedge t'' = t''']]] \wedge t' \subseteq t \rightarrow P(t')]$

From that example alone, it is clear that t^* is not bound in the same way as indexical pronouns are bound in the nominal domain.

We can also see that habitual anchoring is different from binding if we observe how temporal adverbials are constrained under temporal quantifiers in a not-habitual environment. Example (37) shows the quantifier *kašəŋ šos* ‘every time’, which also occurs in habitual environments. However, the number of iterations is explicitly specified in the context of (37). In the restrictor of *kašəŋ šos* ‘every time’, there are two events of bringing a younger brother to the *kərt* (nomad camp), both of which are located in the past and both of which overlap with him crying. Contrary to what we observe in habitual sentences, NFIN.PST is felicitous and NFIN.NPST is not. Moreover, consultants observe that NFIN.NPST gives rise to a different interpretation, namely, that the brother was taken away multiple times and not just twice.

- (37) //My younger brother has been at the kɔrt twice//
 [λu w tɔw-əm-ən / #tɔ-tij-ən] kaʂəŋ ʂos χoλλə-s
 (s)he take-NFIN.PST-LOC / #take-NFIN.NPST-LOC every time cry-PST[3SG]
 ‘When he was taken away, he cried every time’
 Comments on *tɔ-tij-ən* ‘take-NFIN.NPST-LOC’: ‘It means he is frequently taken away’

Example (37) clearly shows that it is not universal quantification over times making NFIN.PST infelicitous in habituals, since in non-habitual sentences involving quantification over times LOC-adverbs exhibit absolute tense just like in episodic contexts. A purely temporal binding-based approach does not seem to grasp the contrast between temporal quantification, as in example (37), and habitual anchoring. With all that considered, habitual anchoring should be explained not by quantification, but by some other property that makes habituals different from regular quantifiers over times.

2.5. Towards Modal Anchoring

There are several approaches to habituality that claim that habitual aspect is not just a plain quantifier over times, but rather a modal operator. Some of them draw apart modal and quantifier-like habituality, see (Hacquard 2006; Boneh & Doron 2008, 2010; Cable 2020). With that in mind, habitual anchoring of LOC-adverbs can be attributed to the same mechanism as other cases of anchoring in modal environments, namely, relative tense in complements of propositional attitudes. Here I will label complements of propositional attitudes as sentential arguments (SAs).

Suppose that habituality cumulatively introduces a habitus — a state at each part of which it is true that a habit holds at that part (Scheiner, 2002). Then, MT of a habitual sentence is a relevant time period at which the habitus takes place. The resulting ordering is illustrated below.

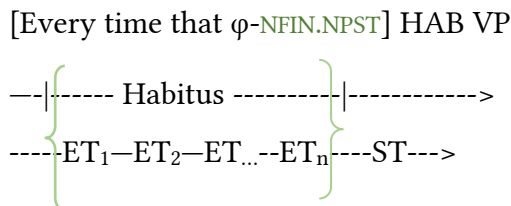


Figure 7. *ET iteration in habituals*

Here, every rotation of ET overlaps with the habitus. The last step necessary to derive the NFIN.NPST marking in LOC-adverbs is to somehow deliver that habitus time to a T-anchor of the LOC-adverb. Essentially, it can be done with the same mechanisms that are used for embedded tense in propositional attitudes. For that, let us examine the temporal properties of Khanty SAs.

Examples (2-3), repeated as (38-39) below show that finite clauses or CP-level gerunds that function as attitude complements are interpreted with relative tense.

- (38) //I called Masha and asked her what Vasya is doing//
 mašaj-en lup-əs [wašaj-en ul / ul-t-aλ /
 m.-POSS.2SG say-PST[3SG] V.-POSS.2SG sleep-NPST[3SG] / sleep-NFIN.NPST-POSS.3SG/
 #ul-əs / #ul-m-aλ]
 #sleep.PST / #sleep-NFIN.PST-POSS.3SG
 ‘Masha said that Vasya was sleeping (when I called)’

 MT=ET ----- ST ---->

- (39) //Today I called Maria and asked her what Vasya has been doing during weekends//
 mašaj-en lup-əs [wašaj-en ul-əs / ul-m-aλ /
 m.-POSS.2SG say-PST[3SG] V.-POSS.2SG sleep-PST[3SG] / sleep-NFIN.PST-POSS.3SG /
 #ul / #ul-t-aλ]
 #sleep.NPST / #sleep-NFIN.NPST-POSS.3SG
 ‘Masha said that Vasya was sleeping (on the weekends)’

 ET ----- MT ----- ST ---->

There is a recurrent observation in the literature on *de re* / *de se* distinctions in tense that if an embedded clause contains a deictic adverbial *then*, then it can only be interpreted with absolute tense, see (Vostrikova, 2019). Then, in case of Khanty, we would also expect non-finite complement clauses to obtain a forced *de re* interpretation with *širən* ‘then’. However, such clauses are simply ungrammatical with any tense, see (40). Therefore, *de re* tense cannot be forced with *then*.

- (40) #χən wašaj-en ow-əl-a seŋk-s-əm im-eλ lup-əs [širən λəw
 when V.-POSS.2SG door-POSS.3SG-DAT knock-PST-1sg wife-POSS.3SG say-PST then he
 ul / ul-əs]
 sleep.NPST[3SG] / sleep-PST[3SG]
 Intended: ‘When I knocked on Vasya’s door, his wife said that he was sleeping (then)’.

Finite complements look as if they were juxtaposed and not subordinated and, indeed, they have some properties that might indicate towards coordination. For example, both finite and non-finite complements in Khanty cannot be nested, see (41) and compare it with nesting in (42), which contains a postposition.

- (41) *mašaj-en [wašaj-en ul-t-aλ / ul] lup-əs
 m.-POSS.2SG V.-POSS.2SG sleep-NFIN.NPST-POSS.3SG / sleep.NPST say-PST[3SG]
 Intended: ‘Masha said that Vasya was sleeping’
 (42) ^{OK}mašaj-en [wašaj-en ul-t-aλ oləŋən] lup-əs
 m.-POSS.2SG V.-POSS.2SG sleep-NFIN.NPST-POSS.3SG about say-PST[3SG]
 ‘Masha told about Vasya sleeping’

However, I suggest that finite SAs are cases of true embedding, because, when propositional attitudes are embedded in counterfactual optatives, only the matrix clause receives counterfactual PST-marking, see ME on that. Taking that into account, I will proceed to analyze finite SAs as embedded clauses and relative tense in these as indexiphoric binding and not anaphoric coreference with the coordinated matrix clause.

- (43) ank-εm λəλən lup-əs ki [ma komputər-ən λət-λ-aj-əm]
 mother-POSS.1SG IRR say-PST[3SG] if I PC-LOC buy-NPST-PASS-1SG
 ‘I wish Mom would buy me a PC’

I suggest to generalize habitual anchoring of gerunds and relative tense in gerunds in SAs in a following manner. Non-finite tense in LOC-adverbs locates ET in relation to the time of habitus and non-finite tense in SAs locates ET in relation to MT of the attitude predicate.

Habitual anchoring:

ST ... [HAB(t_1) [t^* ($=t_1$) NFIN.NPST φ]_{GERUND}]

Propositional attitudes:

ST ... [hear(t_1) [t^* ($=t_1$) NFIN.(N)PST φ]_{GERUND}]

Figure 8. Schemata for t^* shifting in gerunds

We could suggest that t^* refers not to a temporal variable, but rather to a world-time pair. Then, the contrast between habitual anchoring and absence of anchoring with temporal quantifiers could be derived by stating that modals quantify over world-time pairs, and not simply times, and therefore only modals can bind t^* . This approach would introduce structural complications, as it would require both the temporal domain and the domain of worlds to be closed in the same projection. The approach that I introduce below does not bring about such heavy structural stipulations and makes additional predictions regarding the different indexical properties of tense vs temporal indexical adverbs like *tomorrow*.

I suggest that the answer to the anchoring puzzle might come from the indexical(-ish) nature of the t^* anchor. By suggesting that the relative interpretations of embedded tense are similar with indexical shift related phenomena, I make a claim similar to the one made by Schlenker (1999, 2003), who suggests that shifted tense, like other indexical shifts, can be accounted for via monster-operators, that shift embedded indexicals, and quantification over contexts in propositional attitudes. However, his monster-like approach to tense does not seem to work here, since temporal anchoring in Khanty extends beyond propositional attitudes and it would be strange to suggest that habituais quantify over contexts. Next section aims to show how tenses in Khanty are similar to indexicals and different from them, using indexical temporal adverbials like *tomorrow*. Building on that comparison, I will suggest that Khanty non-finite tenses can be plausibly analyzed as indexicals in terms of (Deal 2018; 2020).

3. (Quasi-)Indexicality

Indexical shift is observed in modal environments and is most extensively studied with propositional attitudes. It encompasses cases where indexical expressions are evaluated in local modal environments and not in a general context of utterance. As an illustration, consider (44), containing a sentence in Zazaki from (Anand & Nevins, 2004). Here the first person pronoun can refer both to the original speaker and to Hesên, who is a subject of verb of saying. Shift can also be mandatory and can affect indexical items outside of the nominal domain, see (45). Here the temporal adverb *vizeri* ‘yesterday’ can only be interpreted relatively to the time of H. speaking, not the original Speech Time.

- (44) Hesênij (mîk-ra) va kε εzj/k dεwletia
Hesên.OBL (I.OBL-to) said that I rich.be-PRES
‘Hesên said that {I am, Hesên is} rich.’ (Anand & Nevins, 2004)
- (45) Hefte nayeraraver, H. mî-ra va kε o **vizeri** Rojda paci kerd
week ago H. me-at said that he yesterday Rojda kiss did
‘A week ago, H. told me that he kissed Rojda {8 days ago, #yesterday}.’
(Anand & Nevins, 2004)

Since indexical shift and similar phenomena is usually associated with propositional attitudes, it is not quite clear whether it can be triggered by habituals. However, I suggest there is robust evidence that habitual and generic contexts do in fact affect indexical pronouns cross-linguistically. See examples below, which show how a second person indexical loses its referential meaning in generic sentences. The 2SG pronoun here does not refer to an addressee, but rather to some indefinite generic individual. Generic uses of second person pronouns are quite widespread — Siewierska (2004) reports on a minimum of 7 such languages with Gruber (2011) adding another 11.

- (46) In those days, you always/usually/rarely lived to be 60. (Malamud, 2012)
- (47) Damals musstest du vorsichtig sein.
then must.2SG you.SG careful be
‘In those days, you had to be careful’ (Malamud, 2012)

This data points out that habitual or generic sentences can make indexicals non-deictic in one way or another. If we would assume that both propositional and habitual anchoring of tense are cases where t^* undergoes indexical shift, then both phenomena could be explained by a single mechanism. However, there is a problematic detail — Khanty disallows indexical shift of any kind, including temporal indexicals.

Let us examine adverbs *tāmχātəl* ‘today’ and *χālewət* ‘tomorrow’. First, it is necessary to establish that these are proper indexicals and therefore are not bound by quantifiers. Example (48) shows an adverb *tāmχātəl* ‘today’ in a temporal clause under habitual quantification. Note

that the tense on a gerund in the *before*-clause is anchored. On the contrary, the adverb *tāmχātəλ* cannot appear in this environment and cannot receive a bound reading. The same is shown for *χālewət* ‘tomorrow’ in example (49) below.

- (48) *kašəŋ* *śos* [(#*tāmχātəλ*)*kāti-λ-am* *lapə-t-əm* *jəpijən*] *kiməλt* *χatəλ*
 every time #today cat-PL-POSS.1SG feed-NFIN.NPST-POSS.1SG after second day
juxət-s-ət
 come-PST-3PL
 ‘Every time (#today) after I fed my cats, they returned the following day’
Intended: ‘Every time, after I fed my cats at one day, they returned the following day’

- (49) *kašəŋ* *śos* [*mišaj-en* *ləmətsux-λ-aλ* *nuχ* *pəs-t-əm* *jəpijən*]
 every time [M.-POSS.2SG clothes-PL-POSS.3SG up wash-NFIN.NPST-POSS.1SG after]
imuxti / #*χālewət* *χələŋ-a* *wər-λ-əλλe*
 immediately / #tomorrow dirty-DAT do-NPST-3SG>NSG
 ‘Every time I wash Misha’s clothes, he gets him dirty right away / #tomorrow’
Intended: ‘Every time I wash Misha’s clothes, he gets him dirty the next day’

In SAs, temporal adverbs cannot shift as well. See (50), where *tāmχātəλ* ‘today’ can only be interpreted in relation to ST and never in relation to MT. In a similar fashion, embedded *χālewət* ‘tomorrow’ is always absolute and never shifted, cf. (51).

- (50) [*liw* {*śi* *χātəλ* / #*tāmχātəλ*} *swadba* *wər-t-eλ*] *ma* *wə-s-əm*
 3PL DEM day / #today wedding do-NFIN.NPST-POSS.3PL I know-PST-1SG>SG
təp *məša* *ji-s-əm* *ən* *juxət-s-əm*
 only fall_ill-PST-1SG NEG come-PST-1SG
 ‘I knew they had a wedding that day, but I fell ill and didn’t attend’
Comment: ‘*tāmχātəλ* means today-today, on the day I am talking to you’

- (51) *ma* *radiwa* *χuwat* *χəl-λ’ə-s-əm* {*%χālewət* / *pa* *χātəλ*} *jert-a*
 I radio through hear-plac-PST-1SG %tomorrow / other day rain-DAT
ji-t-aλ
 go-NFIN.NPST-POSS.3SG
 ‘I heard on the radio that I would rain the next day / #tomorrow’
 [But it didn’t rain the next day, and the day after that]

It has been observed that indexicals of a particular class should either all be shifted or remain intact when embedded under a propositional attitude (Anand & Nevins, 2004; Rice, 1986). This pattern is defined as the Shift Together constraint in (52).

- (52) **Shift Together** (Deal, 2020)

If one indexical of class Ψ picks up reference from context *c*, then all indexicals of class Ψ within the same minimal attitude complement must also pick up reference from context *c*.

Note that Shift Together holds for Khanty tense – when a sentence containing a LOC-adverb is embedded under a propositional attitude, both the complement clause and a LOC-adverb are temporally anchored. In example (53), finite past in the complement clause and

NFIN.PST in the LOC-adverb get interpreted in relation to MT, i.e. as preceding Sister's arrival and not ST, although both tenses are interpreted as absolute outside of modal environments.

- (53) //Sis will come home soon! Let's pretend we're asleep!//

up-ew šiwaλ-əλλe ki [[λuw juχət-m-aλ-ən]
 older.sister-POSS.1SG see.NPST-3SG>SG if (s)he come-NFIN.PST-POSS.3SG-LOC
 mən̩ ǎnt ul-s-əw pa ńawšɛmə-s-əw] aŋk-ew-a lup-əλ
 we.PL NEG sleep-PST-1PL add joke-PST-1PL mother-POSS.1PL-DAT say-NPST[3SG]
 ‘If she sees us, she will tell Mom [that [when she came home (PST)] we weren’t asleep and
 were goofing around (PST)]’

ST ----- ET ----- MT --->

ET-LOC

Now, temporal adverbials and a t^* anchor both belong to the domain of time. Therefore, we expect them to obey the Shift Together constraint. As shown below, that is clearly not the case.

	temporal adverbs	non-finite tense
Propositional attitudes	-	shift
Habituals	-	shift

Table 3. Separate shift in the temporal domain

Such a discrepancy makes it clear that temporal adverbials and tense itself have different properties, and either they belong to different indexical classes, or tense is not a proper indexical at all. I argue for the latter and suggest that the temporal anchor t^* is not an indexical, but rather an indexiphor in terms of (Deal, 2018). In the following section I describe the key idea behind the analysis. Its formal implementation is given in Section 5.

4. Indexiphoric tense

Indexiphoricity, as introduced in (Deal, 2018, 2020), covers shift-like behavior of verbal inflection, namely, first and second-person agreement markers. Deal (2020) cites the following examples from Donno So concerning the 1sg agreement marker glossed as -N. In matrix clauses, -N looks like plain first-person agreement.

- (54) a. (Mi) boje-m

1SG go-N
'I'm going'

- b. (Wo) boje

3SG go
'He/She is going' (Culy, 1994, 122 via Deal 2022)

When used in an attitude report, -N agreement signals that the subject of an embedded clause coincides with the attitude holder. Therefore, it acts not as first-person inflection, but rather as “logophoric inflection”.

- (55) Oumar [ma jembɔ paza boli] miñ tagi
 Oumar [1SG sack.DEF drop left] 1SG.OBJ informed
 ‘Oumar informed me that I had left without the sack.’
- (56) Oumar [inyemɛ jembɔ paza bolu-m] miñ tagi
 Oumar [LOG sack.DEF drop left-N] 1SG.OBJ informed
 ‘Oumar_j informed me that he_j had left without the sack.’
- (Culy, 1994, 123 via Deal 2022)

The -N agreement can be analyzed as undergoing obligatory evidential shift. What is strange, however, is that first-person indexical pronouns do not shift in Donno So embedded environments. Following the available evidence, Deal suggests that -N is not a proper indexical, but an indexiphor.

When defining the shifting operators, Anand and Nevins (2004) modify context-dependence using two sets of parameters – context parameter and index parameter, illustrated in (57).

- (57) $\llbracket \alpha \rrbracket^{\text{context}, \text{index}}$
 $\llbracket \text{say } \alpha \rrbracket^{c,i} = \lambda x_e. \forall j \text{ compatible with what } x \text{ says in } i. \llbracket \alpha \rrbracket^{c,j}$ (Anand & Nevins, 2004)

Both context and index are tuples of variables, for example <Speaker, Addressee, Now, World>. There might be more variables, e.g. Korotkova (2016) introduces an Origo parameter specific to evidentials.

The index parameter is affected by modal quantification and the context parameter is not. Therefore, variables that are evaluated according to the index parameter, e.g. world arguments, shift in modal environments. In contrast, indexicals like *I* or *yesterday*, which use the data from the *context* parameter, are typically left unaffected. In matrix clauses *index* is equal to *context*.

For the Donno So N-agreement, Deal suggests that it is interpreted not in relation to the *context* parameter, but in relation to the *index* parameter. Hence it is always shifted in modal environments without additional shifting operators.

- (58) $\llbracket [\text{AUTH-I}] \rrbracket^{c,i,g} = \lambda x. \text{Speaker}(i). x$

I argue that embedded tense in Khanty is subject to the same mechanism and t^* is indexiphoric. It takes the index parameter and returns its Speech Time parameter, which we will label as *now*(*i*). The resulting lexical entry for t^* is shown in (59).

- (59) $\llbracket t^* \rrbracket^{c,i,g} = \text{now}(i)$

As modals quantify over indices, this approach is not essentially different from the binding-based approach. What is important, however, is that it allows us to separate the plain temporal quantifiers like *every time (of N)* which do not constrain non-finite tense, and habitual environments that do. Since modals shift indices of their prejacent, they also shift the world-time parameters that are stored in *index* tuples of <Speaker, Addressee, Now, World, etc.>. This approach also allows us to pinpoint that only those clauses that contain t^* are re-anchored in modal environments in both Khanty and Japanese. In contrast, the clauses that lack absolute tense in indicative sentences and have no t^* pronoun are always interpreted as relative to the closest temporal variable in the local domain.

Recall that Khanty tenses in attitude complements can only have a *de se* interpretation, as was shown in (38-38). Deal (2018) discusses that indexiphors only allow *de se* interpretations and this goes in line with our Khanty data. Deal also mentions a *De Re Blocking Effect*, which states that indexiphors in scope of the binding OP cannot be c-commanded by indexical items from the same domain.

(60) **De Re Blocking Effect** (Deal 2018)

All bound *de se* elements must be *de re* free in the domain of their binding OP.

Note that attitude complements in sentences like ‘Masha said that Vasya will come tomorrow’, which contain both indexical adverbials and the t^* anchor, which is supposedly indexiphoric, do not show De Re Blocking, because t^* always c-commands the indexical adverbials and not vice versa. Another pattern, expected from indexiphors is a restriction on locality — we expect that t^* will only be anchored to the nearest c-commanding modal operator. See example (61), where the LOC-adverb that modifies the SA is anchored to the verb of hearing and not to the habitual operator in the matrix clause, hence NFIN.NPST marking is not available.

(61) // What’s the matter with people? //

kašəŋ χətəλtəp χəλ-λ-əmə [mättə joχ-λ-əw pa ši
 every day only hear-NPST-1SG [some people-PL-POSS.1SG again
 jəχa mən-**{m/#t}**-an]
 marry-**{NFIN.PST/#NFIN.NPST}**-3DU]
 ‘Every day I hear that someone got married!’

Curiously, the indexical-indexiphor distinction also reflects the way temporal adverbs and temporal indexicals are interpreted in Free Indirect Discourse (FID) and Narrative Progression (a.k.a. praesens historicum), see (Schlenker, 2004; Stokke, 2013). Take a look at the examples from (Schlenker, 2004), given below. Example (62) illustrates Free Indirect Discourse (FID), i.e. a context where the speaker puts herself into another person’s shoes and narrates from their perspective. Note that while the sentence in (62) is describing events in the past, the

adverb *tomorrow* is felicitous, as it shifts to a viewpoint of that particular time in the past. The verb, on the contrary, is marked for past tense. Therefore, the evaluation time for tense is not shifted in FID, and evaluation time for adverbials is.

- (62) **Tomorrow was** Monday, Monday, the beginning of another schoolweek! (Schlenker, 2004)

Example (63) shows an instance of praesens historicum (Narrative Present). Here, the adverbial *fifty eight years ago to this day* is interpreted in relation to actual time of utterance. However, the verbs are marked for present tense, despite describing events in the past. Here the evaluation time for tense is shifted and evaluation time for adverbials remains at ST.

- (63) **Fifty eight years ago to this day**, on January 22, 1944, just as the Americans **are** about to invade Europe, the Germans **attack** Vercors. (Schlenker, 2004)

Schlenker suggests that this divergence can be accounted for by assuming two separate contextual parameters: Context of Utterance and Context of Thought. According to Schlenker, tenses and pronouns are evaluated in relation to the Context of Utterance, and demonstratives and temporal adverbials are evaluated in relation to Context of Thought.

	temporal adverbs	finite tense
Free Indirect Discourse	shift	-
Praesens Historicum	-	shift
Evaluation Time	context (Context of Thought)	index (Context of Utterance)

Table 4. *Separate shift in the temporal domain*

Eckardt (2014) analyses FID as evaluation relative to two contextual parameters $\langle C, c \rangle$ instead of one, which is similar to our approach with adverbs. Abrusán (2021) observes that Narrative Present (subsumed under a category of Protagonist Projection) is embeddable, unlike FID, and therefore can scope lower than the speech act projections. This might serve as a bridge towards analyzing FID as context shift and Narrative Present as index shift. However, in absence of systemic data on FID in Khanty it remains on hand-waving ground.

In the following section I outline the formal analysis of modal anchoring and discuss its predictions.

5. Analysis

5.1. Deriving anchoring in propositional attitudes

From now on, we will use the following superscripts: c for context, i for index and g for the pronominal assignment. To derive index shift we will use Intensional Function application (Heim & Kratzer, 1998), as it was formulated in (Deal, 2020).

(64) **Intensional Function Application** (Deal, 2020)

If α is a branching node and $\{\beta, \gamma\}$ the set of its daughters, then for any context c , index i , and assignment g : if $\llbracket \beta \rrbracket^{c,i,g}$ is a function whose domain contains $\lambda i'. \llbracket \gamma \rrbracket^{c,i',g}$, then $\llbracket \alpha \rrbracket^{c,i,g} = \llbracket \beta \rrbracket^{c,i,g} (\lambda i'. \llbracket \gamma \rrbracket^{c,i',g})$

We will also be using the generalized Predicate Conjunction rule, as given in (Wood, 2015), see below.

(65) **Predicate Conjunction** (Wood, 2015)

If α is a branching node, $\{\beta, \gamma\}$ is the set of α 's daughters, and $\llbracket \beta \rrbracket$ and $\llbracket \gamma \rrbracket$ are both in D_f , f a semantic type which takes n arguments, then $\llbracket \alpha \rrbracket = \lambda(a_1, \dots, a_n). \llbracket \beta \rrbracket(a_1, \dots, a_n) \wedge \llbracket \gamma \rrbracket(a_1, \dots, a_n)$.

Following Deal, I assume that gerunds denote propositions and SAs contain a silent operator CONT that turns a proposition $\llbracket \alpha \rrbracket^{c,i,g}$ into a function that selects for events e such that the content of e is a proposition with indexes assigned relative to the e . The resulting denotation of a SA is given in (66). EXT(e) is a function that returns an external argument of an event and $\tau(e)$ returns the event's temporal trace. Note that below I will use w as a shorthand for $world(i)$ and w' as a shorthand for $world(i')$ accordingly.

(66) $\llbracket SA\alpha \rrbracket^{c,i,g} = \lambda e. \forall i' \in \text{CONT}(e) \llbracket \alpha \rrbracket^{c,i',g}$

$i' \in \text{CONT}(e)$ iff

Speaker(i') is an individual in $w(i')$ that EXT(e) identifies in e as herself;

...

world(i') = w' , w' is a member of content of e ;

now(i') = $\tau(e)$.

With rich semantics of SAs in place, the matrix verb will be reduced to a simple predicate of events, as in (67).

(67) $\llbracket \text{lup-} \rrbracket^{c,i,g} = \lambda e. \text{speaking}(e)$

The composition of sentence (68) is illustrated below in Figure 9 and in (68-70).

(68) //Today I called Masha and asked her what Vasya has been doing during weekends//

mašaj-en lup-əs wašaj-en uλ-əs / uλ-m-aλ /
m.-POSS.2SG say-PST[3SG] V.-POSS.2SG sleep-PST[3SG] / sleep-NFIN.PST-POSS.3SG /
#uλ / #uλ-t-aλ
#sleep.NPST / #sleep-NFIN.NPST-POSS.3SG
'Masha said that Vasya slept'

 ET ----- MT ----- ST ---->

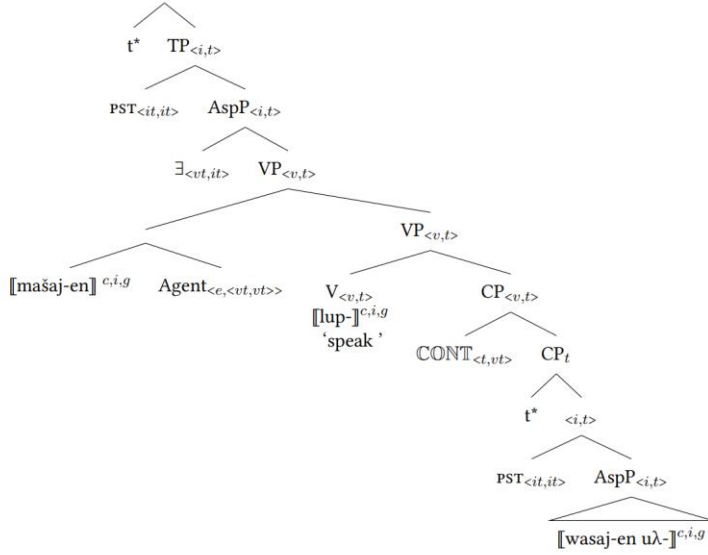


Figure 9. SA composition

- (69) $\llbracket \text{embCP} \rrbracket^{c,i,g} = \lambda e. \forall i' \in \text{CONT}(e) \llbracket \text{wašaj-en u}\lambda\text{-əs} \rrbracket^{c,i',g}$
 $= \lambda e. \forall i' \in \text{CONT}(e) [\exists t. t < \text{now}(i') \wedge \text{wasja_sleep}(t)(w')]$
 $i' \in \text{CONT}(e)$ iff
 speaker(i') is an individual in $w(i')$ that EXT(e) identifies in e as herself;
 ...
 world(i') = w' , w' is a member of content of e ;
 now(i') = $\tau(e)$.
- (70) $\llbracket \text{VP} \rrbracket^{c,i,g} = \lambda e. \text{saying}(e) \wedge \forall i' \in \text{CONT}(e) \llbracket \text{wašaj-en u}\lambda\text{-əs} \rrbracket^{c,i',g}$
- (71) $\llbracket \text{CP} \rrbracket^{c,i,g} = \exists t. t < \mathbf{now(i)} \wedge \exists e [\tau(e) = t \wedge \text{speaking}(e)(w) \wedge \forall i' \in \text{CONT}(e) \llbracket \text{wašaj-en u}\lambda\text{-əs} \rrbracket^{c,i',g}] =$
 $\exists t. t < \mathbf{now(i)} \wedge \exists e [\tau(e) = t \wedge \text{speaking}(e)(w) \wedge \text{Agent}(e) = m \wedge$
 $\forall i' \in \text{CONT}(e) [\exists t'. t' < \mathbf{now(i')} \wedge \text{vasya_sleep}(t')(w')]] =$
 $\exists t. t < \mathbf{now(i)} \wedge \exists e [\tau(e) = t \wedge \text{Agent}(e) = m \wedge \text{speaking}(e)(w) \wedge$
 $\forall i' \in \text{CONT}(e) [\exists t'. t' < t \wedge \text{vasya_sleep}(t')(w')]]$

The resulting denotation states that there exists a time t such that at that time Masha said about Vasya sleeping at some time t' , that is earlier than t . This is completely in line with Vasya sleeping in the past relative to MT and not ST.

5.2. Deriving habitual anchoring

For a habitual operator, I will use a slightly modified version of Ferreira's analysis (2016; 2005). Ferreira suggests that habituals are modals, akin to progressives, but unlike the latter, habituals select for plural and not singular events.

Ferreira assumes that the domain of events contains pluralities, which are mereological sums of other events. Therefore, VP denotations are not limited to singular events, and also include plural ones, in the same way as DPs may denote plural entities. To filter for either

plural or singular events, Ferreira postulates the following operators, which selects events from a given set based on their mereological properties. The *sg* operator selects for atomic events and the *pl* operator selects for homogenous sums of events. A sum of events consists of non-overlapping proper parts which are also elements of a given set *P*.

$$(72) \text{ sg} = \lambda P. \lambda e. \min(e, P)$$

$$\min(e, P) \Leftrightarrow P(e) \wedge \neg \exists e' < e [P(e')]$$
 (Ferreira 2016)

$$(73) \text{ pl} = \lambda P. \lambda e. \text{sum}(e, P)$$

$$\text{sum}(e, P) \Leftrightarrow P(e) \wedge \exists e_1, e_2, \dots, e_n < e [P(e_1) \wedge P(e_2) \wedge \dots \wedge P(e_n) \wedge \otimes(e_1, e_2, \dots, e_n) \wedge e = e_1 \oplus e_2 \oplus \dots \oplus e_n]$$
 (Ferreira 2016)

Given the VP denotation that contains the events in (72a), the *pl* operator brings them down to the ones shown in (72b).

$$(74) \text{ a. } \llbracket \text{VP} \rrbracket = \{e_1, e_2, e_3, e_1 \oplus e_2, e_2 \oplus e_3, e_1 \oplus e_3, e_1 \oplus e_2 \oplus e_3\}$$

$$\text{b. } \text{pl}(\llbracket \text{VP} \rrbracket) = \{e_1 \oplus e_2, e_2 \oplus e_3, e_1 \oplus e_3, e_1 \oplus e_2 \oplus e_3\}$$

The core of Ferreira's proposal is to suggest that habituals are existential quantifiers over plural events and progressives are quantifiers over singular ones.

$$(75) \text{ Progressive: } \exists e : \dots \text{sg}(P)(e) \dots$$

$$\text{Habitual: } \exists e : \dots \text{pl}(P)(e) \dots$$

To account for modal properties of habituality, Ferreira uses Portner's (1998) modal analysis of progressive. Under such an analysis, the progressive imposes an ordering, based on a set of propositions *O*, on a set of worlds in which all the propositions of modal base *M* are true, and chooses the highest-ranking worlds via the function *BEST*. For habituality, *M* is a circumstantial modal base and *O* is an ordering that includes the propositions that must be true in order for the sequence of events to proceed and the habit not to be broken. Although Ferreira's implementation anchors *BEST* to time arguments, here we will follow Portner's original analysis and use events as arguments of *BEST* instead of times. It will allow us to use the function *EXT*(*e*) to derive new parameters for indexes that are being quantified over in the same way as it was done above for SAs. The resulting semantics of the habitual are shown in (76). Note that $\llbracket \varphi \rrbracket^{c,i,g}$ is a predicate of events.

$$(76) \llbracket \text{HAB } \varphi \rrbracket^{c,i,g} = \lambda t. \exists e. [t \subseteq \tau(e) \wedge \forall i' \in \text{BEST}(\lambda i'. \llbracket \varphi \rrbracket^{c,i',g}, M, O, i, e) [\exists e'. \text{PL}(\llbracket \varphi \rrbracket^{c,i',g})(e') \wedge t \subseteq \tau(e')]]$$

$$i' \in \text{BEST}(\lambda i'. \llbracket \varphi \rrbracket^{c,i',g}, M, O, i, e) \text{ iff}$$

$$\text{author}(i') = \text{author}(i)$$

...

$$\text{now}(i') = \tau(e)$$

$$\text{world}(i) \text{ belongs to a set } W = \{w' : w' \in \cap M(\lambda i'. [\varphi]^{c,i,g}, i, e) \wedge \neg \exists w'' [w'' \in \cap M(\lambda i'. [\varphi]^{c,i,g}, i, e) \wedge w'' <_{o(\lambda i'. [\varphi]^{c,i,g}, w, t)} w']]\}$$

Another modification that should make Ferreira's analysis suitable for Khanty concerns temporal adverbials. To include quantificational adverbs into the architecture of habituals, Ferreira proposes that quantificational adverbs like *always* attach below the habitual operator and may take temporal adverbial clauses as their restrictors. In Ferreira's system, the habitual or progressive operator serves as a border between the domain of events and domain of times. Therefore, quantificational adverbs, that are located below HAB, take predicates events as both of their arguments. It is assumed that temporal adverbial clauses contain not a relativized temporal argument, but rather a relativized event argument, and therefore they function as predicated of events and not predicates of times.

However, the assumption that temporal clauses are predicates of events seems problematic in light of the following data. Khanty temporal clauses not only serve as restrictors of quantificational adverbials, but also modify the time of habitus. If temporal clauses are predicates of events, they cannot modify the temporal argument of the habitual. See (77), where the outer LOC-adverb [*When my children were younger...*] is marked with NFIN.PST, since the described state is located in the past and the children are not young at ST. The outer LOC-adverb modifies the temporal argument of the habitual. The inner LOC-adverb [*when I baked them bread*] is habitually anchored and marked with NFIN.NPST. It restricts the repeating events that form part of the habitus and, therefore, appears in the scope of HAB.

- (77) [náwɾɛm-λ-am aja-šək wəλ-**{m/#t}**-ɛλ-ən]
 child-PL-POSS.1SG small-ATT be-**{NFIN.PST/#NFIN.NPST}**-POSS.3SG-LOC
 [náń wɛɾ-**{#m/t}**-ɛm-ən] amtət'λ'ə-s-ət
 bread make-**{#NFIN.PST/NFIN.NPST}**-POSS.1SG-LOC rejoice-PST-3PL
 'When my children were younger, they were happy when I baked bread'

To make temporal modification of HAB possible, we could suggest that it receives event arguments instead of times and then introduce an existential closure over the event domain higher in the structure. However, compared to the alternative proposal, it seems to be an unnecessary complication which does not make any additional predictions.

Giurgea & Stoicescu (2022) observe that single-event imperfectives in Romanian cannot be modified by event time localizing temporal adverbials, unlike perfectives and habituais. To account for that, they suggest that an operator `BOUNDED` turns predicates of events into non-homogenous predicates of tenses. The quantificational adverbs, that are located just below `HAB`, take predicates of times as restrictor and scope and results in a predicate of events, which which feeds the `HAB` operator. The resulting structure together with their denotation of `BOUNDED` is shown below.

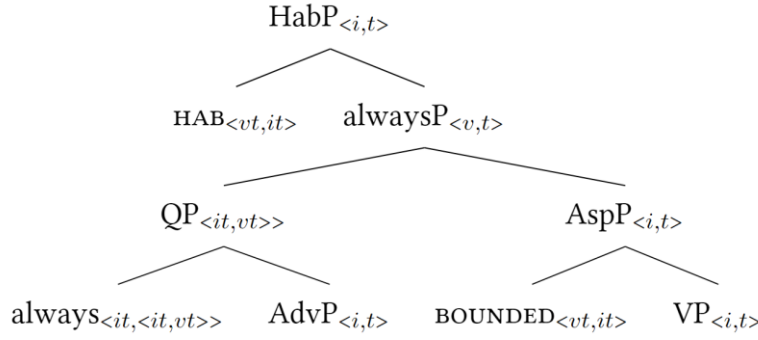


Figure 10. Domains of events and domains of tenses below HAB

$$(78) \llbracket \text{BOUNDED} \rrbracket^{c,i,g} = \lambda P_{\langle v,t \rangle} \lambda t \exists e [P(e) \wedge t = \tau(e)]$$

I will assign the temporal quantifier *isa* ‘always’ the denotation in (79). Here θ is a predicate that is true of two times if they stand in a contextually chosen relation – inclusion, succession etc.

$$(79) \llbracket \text{isa} \rrbracket^{c,i,g} = \lambda Q_{\langle i,t \rangle} \lambda P_{\langle i,t \rangle} \lambda E. \forall t [t \subseteq \tau(E) \wedge Q(t) \rightarrow \exists t' [P(t') \wedge \theta(t, t')]]$$

The derivation of the sentence (80) is given below. Note that, in order to shorten the denotations, I will apply temporal arguments directly to AspPs without introducing lower aspects and components like BOUNDED into the structure.

- (80) ma juχi juχət-{***m/t**}-εm-ən amp-εm ma χuś-am-a
 I home come-{***NFIN.PST/NFIN.NPST**}-POSS.1SG-LOC dog-POSS.1SG I toward-POSS.1SG-DAT
 isa jeśaλt χəχλ-əs
 always to run-PST[3SG]
 ‘When I came back home, our dog would always wait for me’
 //Now she is very old and doesn’t wait for anyone//

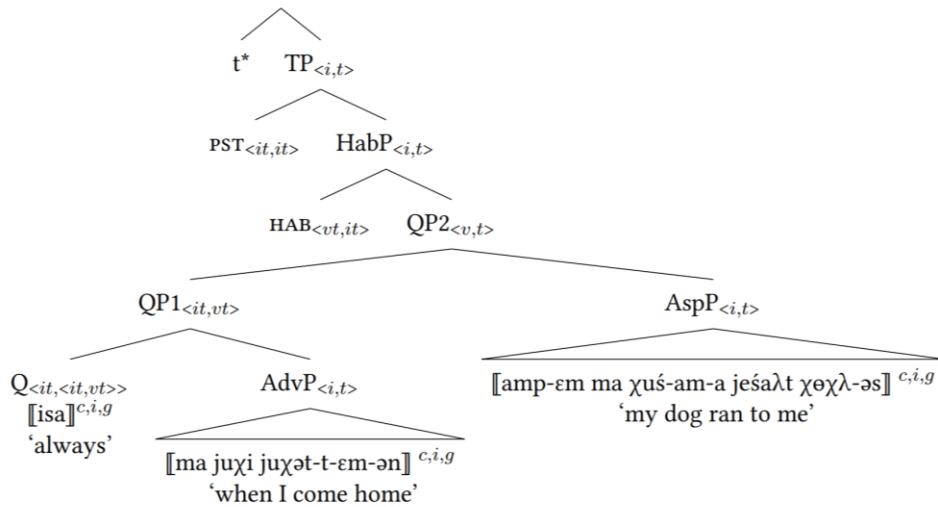


Figure 11. Structure of sentence (80). Habitual anchoring

$$(81) \text{ a. } \llbracket t_1 \text{ amp-ew-ən } \lambda \text{aw}\lambda\text{-aj-əw} \rrbracket^{c,i,g} = \lambda t. \text{dog_wait_for}(g(1))(t)(w)$$

- b. $\llbracket isa \rrbracket^{c,i,g} = \lambda Q_{<i,t>} \lambda P_{<i,t>} \lambda E. \forall t [t \subseteq \tau(E) \wedge Q(t) \rightarrow \exists t' [P(t') \wedge \theta(t,t')]]$
- c. $\llbracket ma \text{ ju}\chi i \text{ ju}\chi\acute{e}t\text{-}t\text{-}\epsilon m\text{-}\acute{e}n \rrbracket^{c,i,g} = \lambda t. \exists t' [t' \geq \text{now}(i) \wedge \text{come_home}(\text{Speaker}(c))(t')(w) \wedge t = t']$
- d. $\llbracket QP_1 \rrbracket^{c,i,g} = \lambda P_{<i,t>} \lambda E. \forall t [t \subseteq \tau(E) \wedge \exists t'. t' \geq \text{now}(i) \wedge \text{come_home}(\text{Speaker}(c))(t')(w) \wedge t = t'] \rightarrow \exists t'' [P(t'') \wedge \theta(t,t'')]$
- e. $\llbracket QP_2 \rrbracket^{c,i,g} = \lambda E. \forall t [t \subseteq \tau(E) \wedge \exists t' [t' \geq \text{now}(i) \wedge \text{come_home}(\text{Speaker}(c))(t')(w) \wedge t = t'] \rightarrow \exists t'' [\text{dog_wait_for}(\text{Speaker}(c))(t'')(w) \wedge \theta(t,t'')]]$
- f. $\llbracket \text{HabP} \rrbracket^{c,i,g} = \lambda t. \exists e. [t \subseteq \tau(e) \wedge \forall i' \in \text{BEST}(\lambda i'. \llbracket isaP \rrbracket^{c,i',g}, M, O, i, e) [\exists e'. \text{PL}(\lambda E. \forall t' [t' \subseteq \tau(E) \wedge \exists t'' [t'' \geq \text{now}(i') \wedge \text{come_home}(\text{Speaker}(c))(t'')(w') \wedge t' = t''] \rightarrow \exists t''' [\text{dog_wait_for}(\text{Speaker}(c))(t''')(w') \wedge \theta(t',t''')]]) (e') \wedge t \subseteq \tau(e')]]$
- =
- $\lambda t. \exists e. [t \subseteq \tau(e) \wedge \forall i' \in \text{BEST}(\lambda i'. \llbracket isaP \rrbracket^{c,i',g}, M, O, i, e) [\exists e'. \text{sum}(e', \lambda E. \forall t' [t' \subseteq \tau(E) \wedge \exists t'' [t'' \geq \tau(e) \wedge \text{come_home}(\text{Speaker}(c))(t'')(w') \wedge t' = t''] \rightarrow \exists t''' [\text{dog_wait_for}(\text{Speaker}(c))(t''')(w') \wedge \theta(t',t''')]]) \wedge t \subseteq \tau(e')]]$

Now let us examine which events are asserted to exist in modally assessable worlds. The habitual states that there exists a plural event e' , such that every subtime t' of the runtime of e' is such that at t' Speaker came home and **the left boundary of t' is not earlier than the left boundary of runtime of e'** , is the time of dog waiting for the Speaker and t is included into the runtime of e' . Since t^* has been re-anchored to $\tau(e)$, at every iteration the time of Speaker coming home is required to begin not earlier than the time of habitus.

Suppose we had the NFIN.PST marking on the LOC-clause. Then we would get the semantics in (83). Here the habitual states that there exists a plural e' , such that every subtime t' of the runtime of e' is such that at t' Speaker came home and **the right boundary of t' is earlier than the left boundary of runtime of e'** , is the time of dog waiting for the Speaker and t is included into the runtime of e' . Here every iteration of the event denoted by the LOC-adverb is required to be located earlier than the beginning of the habit. This would result in an empty restrictor for the *isa* quantifier, because time of every iteration is required to belong to the habitus time by the quantifier itself ($t' \subseteq \tau(E)$) and simultaneously required to be earlier than the beginning of habitus by the NFIN.PST marking on the LOC-adverb. Since the universal quantifier inside the habitual has an empty restrictor, the proposition with $\llbracket \# \text{HabP} \rrbracket^{c,i,g}$ would be trivially true. I suggest that ungrammaticality of NFIN.PST marking in LOC-adverbs under habituals arises because of this triviality. I will not discuss the precise mechanism, however see (Chierchia, 2021; Del Pinal, 2019) for discussion.

- (83) a. $\llbracket ma \text{ ju}\chi i \text{ ju}\chi\acute{e}t\text{-}m\text{-}\epsilon m\text{-}\acute{e}n \rrbracket^{c,i,g} = \lambda t. \exists t' [t' \prec \text{now}(i) \wedge \text{come_home}(\text{Speaker}(c))(t')(w) \wedge t = t']$

$$\begin{aligned}
\text{b. } \llbracket \# \text{HabP} \rrbracket^{c,i,g} &= \lambda t. \exists e. [\forall i' \in \text{BEST}(\lambda i'. \llbracket \text{isaP} \rrbracket^{c,i',g}, M, O, i, e) \\
&\quad [\exists e'. \text{sum}(e', \lambda E. \forall t' [\\
&\quad t' \subseteq \tau(E) \wedge \exists t'' [t'' \prec \tau(e) \wedge \text{come_home}(\text{Speaker}(c))(t'')(w') \wedge t' = t''] \rightarrow \\
&\quad \exists t''' [\text{dog_wait_for}(\text{Speaker}(c))(t''')(w') \wedge \theta(t', t''')]]] \wedge t \subseteq \tau(e')]]
\end{aligned}$$

6. Problematic data

Above I have suggested that all CP-level structures contain the t^* temporal anchor, which refers to ST in non-modal contexts and shifts under modals. In this section I review some data that may seem problematic for such an approach and argue that it is, in fact, not problematic.

6.1. High loc-adverbs

LOC-adverbs may appear in clearly habitual contexts and yet do not exhibit NFIN.NPST marking which is expected in habitually anchored LOC-adverbs. See (84), where LOC-adverb is marked with NFIN.PST. Note that in order to receive NFIN.PST marking a LOC-adverb in habituals should also contain a frequentative (FREQ) marker.

- (84) [ma juχi juχət-ijəλ-**m**-em-ən] amp-ew mǎnt χot χonəŋ
 I home come-FREQ-NFIN.PST-POSS.1SG-LOC dog-POSS.1SG I.ACC house near
 λawət-s-əλλe
 guard-PST-3SG>SG
 ‘When I came home, our dog would always wait for me near the house’
 // Now she is old and doesn’t wait for anybody //

Khanty frequentative marker has iterative and habitual meanings (Tulubyeveva, 2022; Yerbalanov, 2021). I suggest that FREQ-marked LOC-adverbs do not appear in scope of a habitual operator. They have an independent habitual or iterative operator inside the non-finite clause that forms a state of regular actions, in case of (84) it is a state of regularly coming home. Therefore, FREQ-marked LOC-adverbs select continuous time periods at which the iterative state holds instead of episodic instances of the prejacent. With that, they merge above the habitual and modify its time argument. The resulting predicate is true of time periods at which both the habitus of the matrix clause is true and the habitus of the LOC-adverb: ‘When a state of regular A holds, a state of regular B holds’. Compare Figure 12 below, where the LOC-adverb modifies the habitual predicate and Figure 11, where the LOC-adverb appears below HAB.

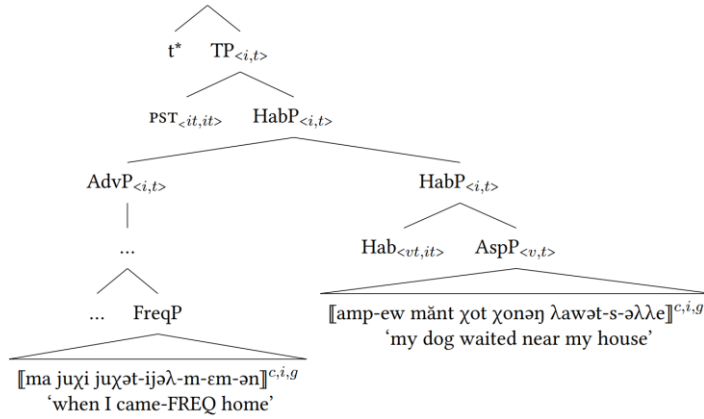


Figure 12. LOC-adverbs modifying the habitual

6.2. Other temporal adverbial clauses

An anonymous reviewer notes that according to Muravyev (2021, 2022) temporal adverbs with postpositions *kutən* and *sa*, that also contain gerunds, are interpreted with relative non-finite tense, see (85). Here we will investigate clauses with *kutən* ‘between’ and show obligatory NFIN.NPST marking in those can be predicted by the current theory.

- (85) mašaj-en urok-λ-aλ wər-{t/#m}-aλ kutən
M.-POSS.2SG homework-PL-POSS.3SG do-{NFIN.NPST/#NFIN.PST}-POSS.3SG between
iλ wujəm-s-a
down fall.asleep-PST-pass
‘While Masha was doing homework, she fell asleep’

Note that adverbial clauses with *kutən* also project up to CP, since they accept epistemic adverbs like *jina* ‘really’, see (86). As these clauses contain high projections, we expect them to contain the t^* anchor which triggers absolute interpretations of tense. Then why are adverbials with *kutən* always marked with NFIN.NPST instead of absolute tense?

- (86) ^{OK}[puχ-εm **jina** urok-λ-aλ wər-t-aλ kutən]
son-POSS.1SG really homework-PL-POSS.3SG do-NFIN.NPST-POSS.3SG between
λəw ješa λ’upiλ-əs
(s)he slightly whisper-PST[3SG]
‘While my son was really working on his homework, he was mumbling’
//And when he started pretending to work, he went quiet//

Unlike LOC-adverbs, these adverbial clauses only accept imperfective embedded clauses. However, it seems like this property alone is not responsible for the obligatory NFIN.NPST marking that we see in (85) for *kutən*. Clauses with postposition *mār* ‘during’, which also require imperfective prejacent, are marked for absolute tense, just as expected from CP-level adverbials. See (87), where the temporal clause is marked with NFIN.PST, since Masha is doing her homework earlier than ST.

- (87) mašaj-en urok-λ-aλ wər-{m/#t}-aλ⁵ mār
 M.-POSS.2SG homework-PL-POSS.3SG do-{NFIN.PST/#NFIN.NPST}-POSS.3SG during
 iλ wujəm-s-a
 down fall.asleep-PST-pass
 ‘While Masha was doing homework, she fell asleep’

If it is not due to obligatory imperfectivity of non-finite complements, then why is tense in *kutən*-adverbs not absolute, despite the fact that they apparently contain a t^* anchor? This mismatch can be solved if we look at particular properties of *kutən*-clauses. These temporal clauses select time periods, at which multiple events *P*, described by the non-finite clause, take place with possible interruptions. *Mār*-clauses, however, only allow for uninterrupted continuous events. That is why example (88) is felicitous with *kutən*, which covers the time of working and taking breaks, and example (89) is infelicitous with *mār*, which only selects times of uninterrupted working. Example (89) results in a contradiction and example (88) does not.

- (88) rəpit-t-ew kutən mən rutśə-s-əw
 work-NFIN.NPST-POSS.1PL between we.PL rest-PST-1PL
 ‘In between working, we rested’
- (89) #rəpit-m-ew mār mən rutśə-s-əw
 work-NFIN.PST-POSS.1PL while we.PL rest-PST-1PL
 Expected: ‘While we worked, we rested’
 Comment: ‘We worked and rested at the same time?’

I suggest that postpositions like *kutən*, unlike *mār*, have aspectual properties of their own and *kutən* creates plural states, similarly to habituais. However, in order to provide a tentative semantic denotation for *kutən*, certain amendments must be made regarding the structure of temporal adverbial clauses. According to the analysis in Section 5, habituais and imperfectives take predicates of events as their arguments. Now note that we analyzed complements of temporal connectors as predicates of times, which are the resulting of relativizing a temporal argument out of a proposition. To construct *kutən*-adverbs as temporal connectors with aspectual content of their own without rewriting the semantics of habituality, we need them to take complement that are predicates of events, rather than of times. This can be fixed by assuming that gerunds in *kutən*-adverbs have a relativized event argument and not a relativized temporal argument. That would turn them into predicates of events of type $\langle v, t \rangle$. If needed, this alteration can be also applied to our analysis of LOC-adverbs above to make the system more consistent. In that case, our semantics of the LOC temporal connectors would change to (90). However, this alteration would not affect the predictions made by our current analysis.

⁵ Adverbial clauses with *mār* ‘during’ can have possessive agreement either on the non-finite form as V-NFIN-POSS *mār*, or on the postposition V-NFIN *mār*-POSS. It does not affect the temporal interpretation.

- (90) a. $\llbracket \text{LOC (when)} \rrbracket = \lambda P \in D_{\langle v, t \rangle}. \lambda t \in D_i. \exists e [P(e) \wedge \tau(e) = t]$
 b. $\llbracket \text{ma ju}\chi\text{i ju}\chi\text{ət-m-εm} \rrbracket^{c,i,g} = \lambda e. \exists t' [t' \prec \text{now}(i) \wedge \text{come_home}(\text{Speaker}(c))(e)(w)]$

I suggest that *kutən* has essentially the same semantics as a habitual, with a different modal base M and ordering source O. Just as a habitual, *kutən* selects a temporal interval which contains several events *P*, however the distance between these events and the required circumstances might be different.

- (91) $\llbracket \text{kutən } \alpha \rrbracket^{c,i,g} = \lambda t. \exists e. [t \subseteq \tau(e) \wedge \forall i' \in \text{BEST}(\lambda i'. \llbracket \alpha \rrbracket^{c,i',g}, M, O, i, e) [\exists e'. \text{PL}(\llbracket \alpha \rrbracket^{c,i',g})(e') \wedge t \subseteq \tau(e')]]]$

$i' \in \text{BEST}(\lambda i'. \llbracket \alpha \rrbracket^{c,i',g}, M, O, i, e)$ iff

author(*i'*) = author(*i*)

...

now(*i'*) = $\tau(e)$

world(*i'*) belongs to a set $W = \{w' : w' \in \cap M(\lambda i'. \llbracket \varphi \rrbracket^{c,i',g}, i, e) \wedge \neg \exists w'' [w'' \in \cap M(\lambda i'. \llbracket \varphi \rrbracket^{c,i',g}, i, e) \wedge w'' <_o(\lambda i'. \llbracket \alpha \rrbracket^{c,i',g}, w, t) w']]\}$

As the t^* anchor in the embedded clause is located in scope of *kutən*, temporal adverbials with *kutən* undergo temporal anchoring in the same way as LOC-adverbials and other temporal clauses with aspectually vacuous connectors undergo anchoring in scope of habituals. Although the analysis of *kutən*-clauses might need to be fleshed out in light of additional data, I suggest that even in its current rough form it shows how NFIN.NPST marking in *kutən*-clauses can be explained away within the system we have built up in this work.

7. Conclusion

In this paper we have discussed the system of tense in Kazym Khanty temporal adverbials and sentential arguments. It was shown that Khanty temporal adverbials contain absolute tense which is shifted in habitual contexts and under propositional attitudes. After comparing the shifting of grammatical tense with shifting of indexical temporal adverbials like *tomorrow*, I have suggested an emerging pattern can be explained by assuming that tense or, more precisely, the temporal anchor t^* is not an indexical, but an indexiphor in terms of Deal (2018,2020). Such an analysis provides a new perspective on tense and makes a step towards more comprehensive formal accounts of embedded tense in underdescribed languages.

Appendix I. Glosses

A>B – subject-object verb agreement;

DAT – dative;

FREQ – frequentative;

IRR – irrealis particle;

LOC – locative;
NFIN – non-finite form;
NEG – negation;
NPST – nonpast tense;
NSG – non-singular number;
PASS – passive;
PL – plural number;
POSS – possessive;
PST – past tense.

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